Failure Attribution Guides a Deterministic Symbolic–Neural Critic for Slides

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Abstract

Vision–language models (VLMs) can pass slides with visible defects, yet the inspection result alone does not identify the source of failure. We test whether separating failure mechanisms can guide the choice of inspection method. A structured-geometry oracle distinguishes defects below the tested perceptual threshold, defects suppressed by a whole-rubric prompt, and *reference-assisted* defects that become identifiable only when a clean twin is available. This diagnosis informs a manually specified critic, frozen before evaluation: trusted native geometry goes to symbolic linters, render-only failures to atomic VLM queries, and semantic content to direct neural inspection. Across three vendors, one atomic query improves balanced accuracy by 0.20–0.31 over a budget-favored control using ten repeated whole-rubric samples. On a disjoint nine-class image arm, the frozen critic reaches 0.826 macro balanced accuracy. Its four trusted-native-IR classes reach perfect balanced accuracy. Weaker neural transfer and unresolved reference requests delimit the result.

1 Introduction

Slide- and document-generation agents need critics that can locate layout and content defects before an output is returned or revised. General-purpose vision–language models (VLMs) offer a convenient critic because they can grade a rendered slide directly against a rubric. Their verdicts, however, leave a practical ambiguity. A model may answer "no defect" when a title visibly spills out of its box either because it cannot resolve the overflow or because the prompt fails to elicit a usable signal. The two explanations call for different remedies: better visual evidence in the first case, and a different query or symbolic tool in the second.

The ambiguity extends beyond a single overflow example because slide defects vary in scale and in the evidence needed to identify them. Some defects are *sub-perceptual* under the tested protocol, such as a three-pixel alignment offset or a one-point font mismatch below the models' effective acuity. Changing the prompt, scale, or resolution does not recover them. Other defects are *format-suppressed*: the model detects them when asked one targeted question with required evidence, but misses them when one pointwise call must score the full rubric. A third set is *reference-assisted*. These defects become resolvable when the model can compare the slide with a clean twin, but not from the single slide alone. A generic label such as "visual understanding failure" collapses these outcomes and obscures the appropriate intervention.

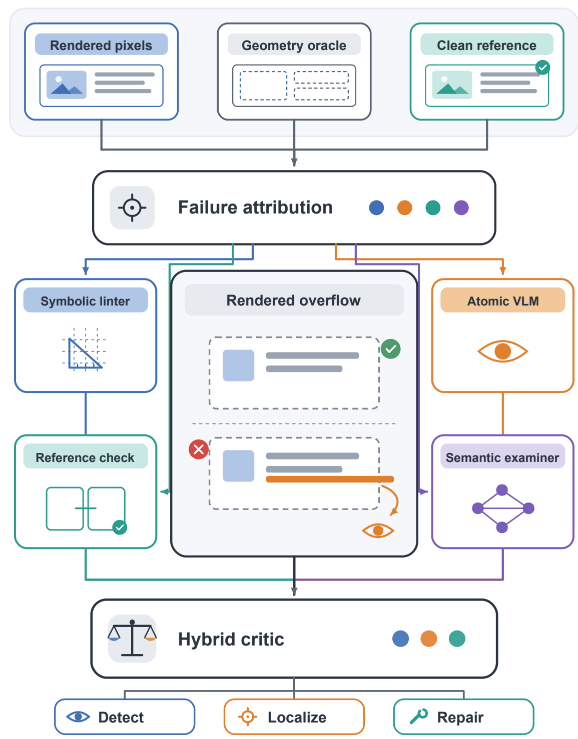

Figure 1: A geometry oracle attributes each failure before guiding a manually specified division of labor among symbolic linting, atomic VLM elicitation, reference requests, and direct semantic inspection.

To separate these outcomes, we use an attribution protocol based on a structured-geometry oracle that is lossless with respect to the declared intermediate representation (IR). The IR contains element bounding boxes, text, and style. We present matched defective and clean slides as an image, the structured oracle, a VLM-generated caption, or the image together with the oracle. We cross these inputs with detection, localization, and repair tasks. Unlike a model-generated caption, the oracle exposes the declared structure without first passing it through another visual model. Success with the oracle after failure on the image therefore operationally identifies a perception bottleneck at the tested interface. Failure with both inputs indicates that changing the neural input alone does not solve the class.

The resulting attribution determines a fixed division of labor (Fig. 1). Symbolic linters inspect declared geometry and terminology. A language model handles text and structural semantics. VLMs inspect perceivable defects under the elicitation format supported by the diagnosis, using either an atomic query or a clean reference. We also construct G7, a render-containment overflow whose declared box is legal although its rendered content crosses the boundary. This class tests the part of the design that a structure-only linter cannot observe. This symbolic–neural critic assumes an IR-owning generation agent. When only third-party pixels are available, the linter lacks its required input and the available critic reduces to the re-elicited VLM (Sec. 7).

The study follows this progression from diagnosis to inspection design. The structured-oracle protocol first separates sub-perceptual, format-suppressed, and reference-assisted failures, including on real CC-licensed layouts (Secs. 4 and 7). Defect-specific elicitation then recovers signals suppressed by a whole-rubric prompt, whereas G1 text overflow and S6 image/text contradiction still require a clean reference (Sec. 5). A repeated-C0 control with a larger sampling budget tests whether compute alone explains the recovery across three vendors. These diagnostic and development results specify a manual critic that is frozen before evaluation on a disjoint nine-class image arm. The Technical Supplement keeps development-stage method selection, full per-class results, and post-hoc executor audits separate.

2 Related work

The VLM perception bottleneck. Prior work separates several sources of visual failure. Chart understanding can be limited either by the vision encoder or by extraction from its representation (Liu et al. 2025b). Direct readouts likewise show that visual information available in an encoder may not be used effectively by the full VLM (Fu et al. 2025). MMVP constructs image pairs that expose systematic shortcomings associated with CLIP-based visual representations (Tong et al. 2024). We study quality inspection of generated artifacts, where declared geometry provides a structured observation of the visual content and the attribution result can be used to choose the critic's inspection method.

Perception/reasoning attribution. Concurrent work uses natural-language descriptions to separate perception from reasoning in abstract visual-reasoning benchmarks (Wang et al. 2025). Our setting instead provides a structured representation of boxes, text, and style, together with exact labels from controlled injection. Unlike a model-written caption, this representation does not pass the evidence through the visual model being diagnosed. The attribution can then inform the downstream inspection method. The LED benchmark also injects empirically grounded structural errors, but it evaluates errors in document-layout-analysis predictions and studies limitations of overlap-based metrics (Heo et al. 2025). We inject errors into rendered slide artifacts and use the resulting diagnosis to compose a critic.

Relative vs. absolute judgement. VLM judges can preserve rankings while producing uncertain absolute scores (Kumar et al. 2026). Pairwise assessment has also improved zero-shot NLG evaluation for some LLMs, although it exhibits positional bias (Liusie, Manakul, and Gales 2024). Protocol choice can introduce other biases as well: pairwise judgments may be more sensitive than absolute scores to distractor features (Tripathi et al. 2025). We use relative and atomic queries as diagnostic interventions rather than assume that either is generally superior. Their class boundary is the relevant result: they recover G1 text overflow and S6 image/text contradiction, but not fine G3 alignment or S3 terminology drift.

Slide generation, evaluation, and reward. Recent slide systems combine generation with evaluation or revision. AutoPresent studies programmatic slide generation and iterative self-refinement (Ge et al. 2025). AeSlides optimizes layout with verifiable aesthetic rewards (Pan et al. 2026), while EvoPresent uses an aesthetic model for scoring, adjustment, and comparative feedback (Liu et al. 2025a). VLM-SlideEval tests element extraction, perturbation sensitivity, and narrative ordering, and argues for calibrated critics in iterative pipelines (Kang, Bao, and Goswami 2025). SlideAudit contributes an expert-derived defect taxonomy and annotated slides for evaluating design critique (Zhang et al. 2025). We use its taxonomy and real slides for transfer evaluation, then audit five published scorers and one closed VLM judge on the constructed render class (Sec. 5.3).

3 Problem setup and defect taxonomy

Task and representation. The critic inspects a single rendered slide, optionally with its declared structure, and must *flag and localize* defects of a fixed set of types. A detection counts only when it fires on a defective slide and stays silent on its matched clean twin. Each slide is represented by an intermediate representation (IR) of typed elements with bounding boxes, text, and style. We inject defects programmatically into the IR and render it, yielding exact labels and matched (defective, clean) pairs. The taxonomy (Table 1) has a *geometry* group G1–G6, a *semantic* group S1–S6, and one constructed render class G7. It is aligned to the expert-derived SlideAudit taxonomy (Zhang et al. 2025) via a bidirectional map (crosswalk in the supplement), and results are reported in its canonical classes. We add G7 to test the linter's blind spot: the declared box is legal (inside the safe margin, no overlap) but the rendered content overflows its container, so the defect exists *only* in pixels. On real agent-generated decks G7 is common, affecting most slides of a Zenodo10K slice and over half of an AutoPresent slice.[1]

Attribution modalities and metric. To attribute a failure, we define four input conditions. Condition A contains only

[1]93-deck Zenodo10K slice: 81/93 decks affected (857/3167 legal boxes); 14-deck AutoPresent slice: 8/14 decks (20/50 legal boxes).

Table 1: Defect taxonomy. Geometry classes with trustworthy declared structure are handled symbolically. Semantic classes require text or visual reasoning, and G7 is constructed to be linter-blind. "Channel" is the inspection engine suggested by the diagnosis (Sec. 4–5).

Group	Class	Operational description	Diagnostic channel
Geometry	G1 text overflow	text exceeds its declared box	VLM + linter
	G2 element overlap	declared element boxes intersect	linter
	G3 alignment offset	element departs from the alignment grid	linter
	G4 font-size inconsistency	font size departs from the deck scale	linter
	G5 brand-color violation	color is off-palette under CIELAB ΔE	linter
	G6 margin violation	element crosses the declared safe margin	linter
Semantic	S1 title/body mismatch	body content contradicts the title	VLM
	S2 narrative-order break	deck sections appear out of logical order	LLM
	S3 terminology inconsistency	a concept's terminology drifts across the deck	term linter
	S4 density violation	slide is substantially over- or under-packed	LLM
	S5 missing logic section	a required step in the argument is absent	LLM
	S6 image/text contradiction	a figure contradicts its caption or surrounding claim	VLM
Render	G7 containment overflow	declared box is legal, but rendered content crosses it	VLM (atomic)

the rendered image. Condition B contains the structured oracle, which preserves element boxes, text, and style from the source IR without loss. Condition B′ contains a VLM-written caption and serves as a lossy control. Condition C combines the image and oracle. Under each condition, we ask the model to detect, localize, and repair the defect. The A-vs-B contrast provides the attribution. If the model fails on the image alone (A) but succeeds given the oracle (B), we call the class *perception-bottlenecked* operationally: the task becomes solvable when the same information is made explicit in structure that is lossless with respect to the declared IR. This diagnosis identifies which engine can solve the class. It does not make a claim about the model's internal representations. If it fails under both, the class is unsolved even at the structured interface, and we assign it to a symbolic method.[2] The defective slide and its matched clean twin form the basic scoring pair: a detection counts only if it fires on the defective member and stays silent on the clean member, so a model cannot score by flagging everything. Balanced-accuracy intervals are component-wise Wilson-derived intervals, formed from the Wilson intervals for recall and specificity. For two-alternative forced choice (2-AFC), the two orderings are order controls rather than independent defect instances. Full intervals and ordering-level results are in the Technical Supplement.

Two linters. A geometry linter detects G1–G6 from coordinates and rules, using the CIELAB color-difference measure ΔE_{2000} rather than RGB distance for brand color. It produces zero false positives on clean slides. A terminology linter handles S3 from an occurrence table plus near-duplicate clustering, image-free. Both are the "detectors of record" selected from the diagnosis.

4 Diagnosis: two kinds of blindness

Pointwise image-only inspection misses fine geometry. Under the conventional setup, one slide receives one rubric score with no reference slide for comparison. We call this *pointwise image-only* inspection. In this setting, G2–G6 sit at chance for 4B and 8B models, and a 30B breaks only G1 overflow. The floor persists under the tested changes. swapping the vision encoder across *five* families (SigLIP2, an LLM-based encoder, InternViT, a native-resolution NaViT, MoonViT) leaves every model at the random line, differing only in abstain-vs-over-flag bias. Raising resolution 1536→2048 changes nothing. Elicitation and magnitude change this result, whereas encoder choice and pixel budget do not. The observed floor is thus a property of the *pointwise* protocol, not a hard capability limit. For declared geometry, the symbolic linter reads coordinates directly and reaches balanced accuracy 0.75–1.0 with zero false positives where the IR is available.

Some apparent perceptual failures depend on elicitation. The same setting hides a second, opposite phenomenon. For G1 overflow and S6 image/text contradiction, pointwise balanced accuracy is 0.50, indistinguishable from the sub-threshold tail, yet a two-alternative forced choice between the defective and clean slide jumps to 1.00: the defect is perceivable, but the single-slide pointwise call does not surface it. We treat G1/S6 as *reference-assisted*: recovery needs the clean twin, not just a renamed single-image question. G3 alignment behaves differently, but only *above a magnitude threshold*: sweeping the injected offset as an internal contrast (one element shifted from its aligned sibling column), the forced choice saturates to 1.00 by 16 px (0.83% of slide width) while the sub-threshold tail ($\leq$ 8 px) stays at chance even side by side. Fine alignment is thus not a class-level capability floor: above threshold it is format-suppressed and recoverable, with a sub-perceptual residue

[2] The oracle's ground-truth labels are stripped from the model's view by a regression-tested de-leaking pass, so B tests perception of structure, not label leakage. B′ is a control, not a second oracle: a caption re-injects the perceptual error it is meant to factor out, so B's advantage is by construction.

Table 2: Attribution-to-intervention correspondence. Each assignment is accompanied by a mismatched-method comparison and its observed boundary. BA denotes balanced accuracy. Results marked "development" informed the manual specification and are not part of the later frozen evaluation.

Diagnosis / classes	Assigned intervention	Matched evidence and mismatch check	Evidential boundary
Declared geometry; G1–G6	native-IR geometry linter	Development BA/precision is 1.000 on G1,G2,G3,G5,G6 ($n_+ = n_- = 150$ each), while pointwise VLM cells are 0.50–0.71.	requires trustworthy native IR
Explicit terminology; S3	terminology linter	BA is 1.000 with zero false positives on 40 clean deck controls, versus 0.69 for the VLM.	deck-level text required
Format-suppressed render overflow; G7	atomic C3 query with evidence	Across three vendors, one C3 call reaches 0.92/0.88/0.83 versus 0.61/0.59/0.63 for ten-draw C0 majority ($\Delta = 0.31/0.29/0.20$).	weak models and frozen transfer remain mixed
Reference-assisted; G1,S6	clean-reference comparison	Two-alternative choice reaches 0.97–1.00, whereas single-image pointwise inspection is 0.50–0.65.	reference must exist and be compared
Page semantics; S1,S4	neural semantic inspection	In-domain examiner BA is 1.000 for both classes, versus 0.778/0.529 for the zero-shot 8B model.	S4 sim-to-real transfer fails
Pixels without native structure	recovered-structure linter	It stays at chance on G1,G3,G4,G5; G2 reaches only 0.54–0.55, with element IoU recall 0.30–0.39.	tested extension fails

only in the small-offset tail. Together, these results reveal a *magnitude-gated* recoverable-versus-sub-perceptual boundary rather than a clean geometric/semantic split, and S3 terminology (below) remains the one defect no elicitation rescues.

On the rebuilt corpus we also now have a human spot-check on the actual pairs cited here, not just the earlier diagnosis runs. Replacing only the stale legacy G3/G5 samples and keeping the other 55 pairs fixed, the v2 pass verifies every sampled defect as visible and every clean twin as clean (73/73, Wilson lower bound 0.95 for both), with source-structure faithfulness 55/55 on the auditable classes. The changed cases match the re-operationalisation: G3 shifts from 0/7 on the old external-reference sample to 8/8 on the corrected within-list misalignment sample, and G5 from 0/7 to 10/10 on the corrected within-list chromatic clash sample. The non-replaced 55 pairs are unchanged.

Terminology remains unresolved. Not every consistency defect is format-suppressed. S3 terminology inconsistency is *not* rescued by forced choice (0.625, heavy position bias): reading the same term across deck pages and spotting a subtle variant is below threshold even side by side. The structured-oracle channel, which feeds the model the full deck text with both variants present, also fails (0.56): with both variants already explicit in the input, this failure cannot be attributed to OCR. It instead reflects cross-page term matching that does not survive the yes/no framing. The image channel is consistent with fine-grained glyph discrimination and the oracle channel fails even with the text in hand, so neither neural method is reliable. We assign S3 to a symbolic terminology linter, which reaches balanced accuracy 1.000 against the VLM's 0.69, a negative result that changes the selected method.

The G6 page-offset boundary depends on magnitude. Re-posing a defect as a within-slide contrast does not rescue every class. It fails outright on G6. We redefine G6 margin as a *page offset*: the whole content block is shifted toward one edge (every element together, so internal alignment is preserved and the defect cannot be misalignment), leaving one margin crowded against (or running off) the edge and the opposite side empty.[3] Across all six VLMs and both pointwise framings the page offset stays at chance (roster-mean C3 0.50). The pairwise test that separates *data artifacts* (S6, which recovers to perfect once the figure is rendered) from genuine blindness leaves G6 on the floor for *every* model. A stress-test addendum identifies the boundary: when the same rigid offset moves content 48 px beyond the page edge, both C0 and C3 detect the defect at 0.958 balanced accuracy (11/12 positives, 12/12 clean controls), although C0 never names G6 while C3 does. The blind spot is genuine over the tested moderate, unclipped range but not magnitude-invariant. Overt clipping restores visibility. Declared geometry is still assigned to the linter because it detects both regimes without relying on this visual threshold.

Template snapping can remove injected defects. Enterprise "snap-to-master" templates re-fit element boxes to a grid before rendering, and we find a template silently *absorbs* 45% of injected geometry defects: overlap, alignment, and margin violations become pixel-identical to their clean twins after snapping, while content-determined overflow survives. This creates a trap for datasets that derive labels from the IR but evaluate rendered images. Section 5.3 quantifies the effect and connects it to reward-model evaluation.

5 A frozen critic guided by diagnosis

The diagnosis shows why a single inspection method is insufficient. Diagnostic experiments and development-stage validation guide a manual correspondence from failure type to inspection method. This correspondence is not a learned router, and it is frozen before the disjoint evaluation. Table 2 tests

[3] The original S6 render omitted the figure element, so the cross-modal contradiction was literally unseeable (recall 0). On the valid-figure corpus the S6 2-AFC reproduces at 0.75–1.00 across all six VLMs, confirming it is a render artifact, not blindness.

Table 3: Frozen nine-class image-arm specification. "Native IR" denotes trusted source structure, not boxes recovered from pixels. A missing terminal verdict is retained as DEFER and scored as a miss.

Classes	Required input	Frozen executor and unresolved behavior
G2,G3,G5,G6	rendered image + native IR	geometry linter operating on trusted source structure
G7	single rendered image	atomic evidence-bearing VLM query with localization
S1	single rendered image	open pointwise VLM query (C0)
S4	single rendered image	direct semantic answer
G1,S6	image + clean reference	request a reference; no comparison verdict $\rightarrow$ DEFER/miss

this correspondence against mismatched methods rather than reporting only successful assignments. The Technical Supplement reports the inspected development validation and its executor mapping separately from the frozen result.

Together these contrasts make the assignment rule operational: use a deterministic checker when the predicate is fully defined in trustworthy source structure. Use an atomic render-aware query when the defect exists only in pixels and whole-rubric elicitation suppresses it. Request a clean comparator when single-slide evidence is insufficient, and reserve neural semantic inspection for predicates not expressible as deterministic geometry or terminology rules. Failure under the required input is retained as a boundary rather than rerouted post hoc.

5.1 Elicitation separates format suppression from unresolved capability limits

Four ways to ask. Holding the model and image fixed, we compare four prompting conditions. C0 is one call that scores every defect type against a rubric. C1 asks for a free-form description before classification. C2 uses a synthetic-twin pair in which the clean reference is re-rendered to a grid. C3 issues one yes/no query per defect type and requires localization of the offending element. The primary comparison is C3 against C0. The model, image, and taxonomy remain fixed. Only the whole-rubric versus atomic evidence-bearing format changes, so a gain under C3 identifies sensitivity to the elicitation format under this controlled comparison.

C3 recovers G7 within the capable subset. We scope the claim to models that can see G7 at all: a model counts as "capable" only if, under C3, it both detects the overflow (balanced accuracy ≥ 0.70 and precision ≥ 0.70 on paired clean controls) and points at the overflowing element. Because selection uses the same test items, this subset is descriptive rather than confirmatory. InternVL-8B and Ovis-9B stay near chance and are excluded. Their result marks a capability boundary rather than a format effect. On the four capable models (three scales, three vendors), C3 recovers what C0 suppresses: balanced accuracy 0.50–$0.79 \rightarrow 0.93$–1.00, precision 0.88–1.00. The effect appears across models. Pairing each slide across the four models, C3 fixes 233 G7 items that C0 missed against 2 reversals (McNemar $\chi^2_1{=}227$, $p < 10^{-50}$).[4] C0's one-call rubric does not include G7, which is a render class outside the original taxonomy. C3 asks about that class directly. The next controls separate the effect of naming the class from the effect of using an atomic query.

Naming and query format have different effects. Is it naming the class that helps, or changing how we ask? Two controls separate this. *C0+* is C0 with G7 added to the rubric's defect list. It retains the one-call format but gives the model a name for the class. *C0_named* instead asks about G7 alone, on its own yes/no call. Adding the name to the one-call rubric (C0+) recovers recall (234 defectives newly flagged, $p{=}7{\times}10^{-71}$), but it reduces specificity: clean slides are over-flagged on 55% of pairs, compared with 2.4% for C3, so balanced accuracy remains lower. Asking about G7 on its own call (C0_named) recovers both, and supplying a clean reference slide adds nothing on top. So for G7 the suppressor is the *one-call format*: under a targeted atomic query, the same model on the same image recovers the detection the whole-taxonomy rubric buries. The boundary is G1/S6: naming alone does *not* help. Those classes need the clean reference slide (pairwise choice $\rightarrow$ 0.97–1.00), a *reference-assisted* effect rather than single-slide format suppression. A clean-vs-clean control rules out a forced-pick artifact (capable models answer "tie" on 92–100% of clean pairs).

The recovery persists against a budget-favored repeated-C0 control. The remaining alternative is that atomic elicitation wins only because it spends more test-time compute than a whole-rubric call. We therefore compare one C3 call with majority vote over ten independent C0 samples on three API-served vendors. C3 improves balanced accuracy by 0.20–0.31 even though repeated C0 receives the larger sampling budget. Extra sampling therefore cannot explain the recovery. These contrasts are included in the paper-wide 85-test Holm family. A focused 24-test analysis is reported only as a descriptive companion in the Technical Supplement.

Naming and format controls in the Technical Supplement show that adding G7 to the whole-rubric list recovers recall but harms specificity, whereas an atomic query recovers both. A clean reference adds no further benefit for this class.

Elicitation depends on model capability and defect class. The best elicitation depends jointly on model capability and defect class. No single prompt covers all cases, so inspection must be assigned per diagnosed failure: the C2 synthetic-twin pairwise rescues declared G1 on capable models (snapping

[4] Per-model $p \leq 1.5 \times 10^{-11}$, all surviving Holm correction over the final paper-wide 85-test family. The two reversals are on Gemma4-31B (C3 recall 88/90). Full table in the supplement.

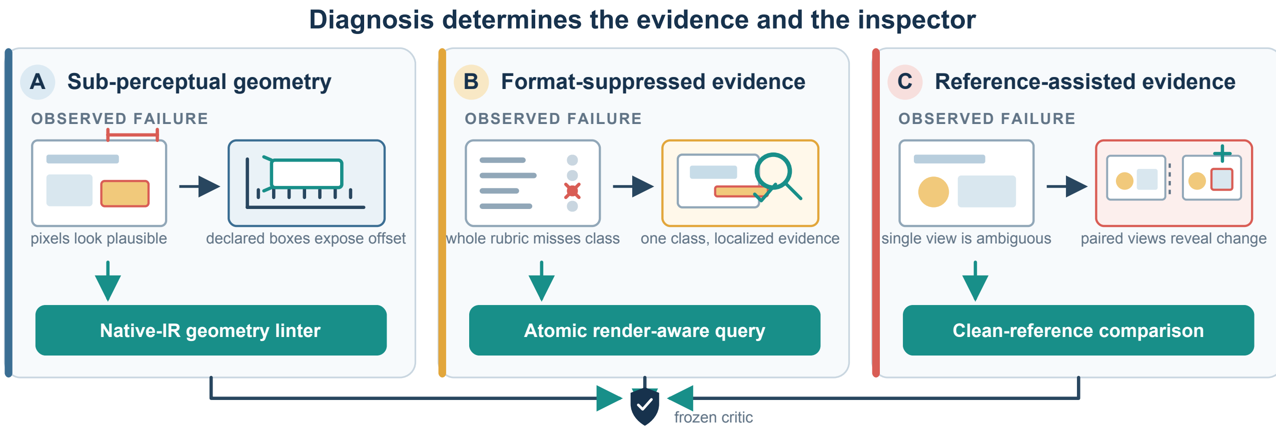

Figure 2: Diagnosis-informed intervention design. Sub-perceptual defects are exposed by declared source geometry. An atomic, localized render query recovers format-suppressed defects, while a clean reference resolves ambiguous single-slide evidence. The diagram summarizes the manually specified correspondence rather than a learned router. Table 2 reports its empirical support and boundaries.

absorbs the overflow and the contrast exposes it), while C1 free-form rescues weak models but collapses on strong ones (precision ≈ 0.50). The format effect is not a synthetic artifact: on human-annotated SlideAudit (no structure) C3 still beats C0 for every class (e.g. brand color, alignment). Absolute geometry on bare real pixels stays modest outside blatant overlap. Real fine geometry needs the linter, hence the structure bare images lack, so on image-only real data the hybrid degrades to VLM-only, a ceiling set by missing structure, not method.

5.2 Frozen critic specification and disjoint evaluation

Table 3 is the executable specification used for the frozen nine-class image arm. It is a manual, diagnosis-informed design rather than a learned router. A missing terminal verdict is DEFER. For evaluation, DEFER is conservatively scored as a miss.

G4/S3 remain linter-owned but are outside this nine-class image arm. The deck-scope classes S2/S5 are also excluded. On the disjoint arm, the frozen critic obtains 0.826 macro balanced accuracy. All four trusted-native-IR linter classes reach 1.000 balanced accuracy. The three direct-neural classes average 0.811. Neither reference request produces a terminal comparison verdict, so G1/S6 are scored as misses. This supports the hybrid division of labor primarily where trustworthy source structure exists, rather than confirming every neural or reference-dependent component. Full per-class precision, recall, localization, and development-stage comparisons are in the Technical Supplement.

A neural transfer boundary. For G7, development performance drops from 0.937 to 0.567 on the frozen arm. We treat this as a transfer failure rather than evidence of universal neural recovery. The available artifacts do not identify a unique cause. Pairwise probes likewise show that G1/S6 can be distinguished when a clean twin is already provided, but the frozen executor does not complete reference request $\to$ comparison $\to$ verdict. The Technical Supplement reports development validation, same-set executor diagnostics, and limited-class follow-ups without treating them as frozen full-system evidence.

5.3 G7: the linter-blind class that specialised rewards also miss

Construction and scorer response. G7 makes the linter's blind spot falsifiable: a declared box that is legal (in-margin, non-overlapping) but whose rendered content overflows its container. The IR carries only legal short text. The overflow lives in the renderer, so the linter and any structure-only model are blind by construction. A self-check confirms zero findings on 90/90 G7 defectives. We then audit five published scorers plus one frontier VLM judge on the same paired slides. The same-backbone comparison isolates the difference most directly: CLIP-IQA detects G7, whereas the LAION aesthetic head on the same CLIP-L/14 backbone is at chance. Skywork, PickScore, and Qwen3.7-Max also detect the class, whereas DocReward does not (full per-scorer table in the Technical Supplement). On the scorers tested, G7 tracks a rendered-quality read-out rather than the training-domain label.

Template snapping removes 45% of injected geometry. The reward audit ties to the template result of Sec. 4. Rendering every injected defective both faithfully and snapped-to-master, 45% of injected geometry defects (100% of overlap, alignment, and margin, and 20% of overflow) become pixel-identical to their clean twin after snapping. Feeding the *snapped* pairs to all five scorers gives preference accuracy 0.00 at a reward gap of exactly 0.000: identical inputs force identical scores. This is a dataset-integrity hazard rather than a reward-model deficiency. It is model-agnostic by construction: any reward model trained or evaluated on snap-rendered images with IR-derived labels inherits these zero-signal pairs. Perturbation fidelity must be checked at the pixel level, not assumed from the IR.

6 A specialist examiner as an in-domain probe

We separately finetune Qwen3-VL-8B with QLoRA on injected supervision to probe whether specialization helps page-semantic inspection. This examiner is not the frozen S4 executor in Table 3, and its intrinsic scores are not substituted into the frozen system result. In-domain, it outperforms larger zero-shot baselines on S1 title/body mismatch and S4 density while abstaining on geometry, as intended. On image-only SlideAudit and a deployed-agent deck, however, the ranking reverses: the specialist does not transfer as a universal bare-pixel inspector. Full training, controls, and results are in the Technical Supplement.

7 External validity

A real pre-sales agent deck exposes distribution shift. On a real 16-slide proposal from a deployed PowerPoint agent, the verifiable defect is 20 unfilled template placeholders ("add a title here") across 7 slides. This semantic-completeness defect is outside the linter's rules. The zero-shot 30B is the best detector (recall 1.0, names the placeholder on 4/7 slides, scores them 0.03 vs. 0.64 for clean), while the finetuned 8B, top of the intrinsic ranking, is among the worst here (0/7), and the linter's apparent recall is spurious geometry noise: on a novel defect the ranking is by general-VLM ability rather than specialist in-distribution quality. The examiner learned a distribution, not universal quality. Feeding the flagged placeholders back to the generator for one revision drives the defect from 20 to 0.

Real layouts reproduce the attribution outcomes. The preceding A/B/C attribution uses synthetic slides. We repeat it on *real* geometry with a declared structural oracle and obtain the oracle without human or model annotation: Zenodo10K (Zheng et al. 2025) ships real CC-licensed `.pptx` with intact element XML, so `python-pptx` extracts a geometry/text/style oracle *lossless with respect to the declared IR* (the IR$\rightarrow$render gap that G7 exploits is exactly what this oracle does *not* capture). From 26 real decks we inject one single-shape defect per slide *in PPTX XML space* (overflow, overlap, alignment, font, margin) and render both the clean and defective one-slide deck with the same renderer (LibreOffice), so the paired pixel difference isolates the injected defect on real pixels (209 paired slides, 5 classes, 3 VLMs from 3 families). The first result is that the §4 "snap absorbs 45%" hazard is *template-specific*: real free-form decks absorb only 7% (render-fidelity 0.93), a positive control. The second result is that all three attribution outcomes appear on real geometry: coarse defects (overflow, overlap) are image-sufficient (A 0.70–0.74, structure adds nothing). A margin bleed is a perception bottleneck that the oracle rescues for weaker VLMs (A $0.59 \rightarrow$ B 0.70), while fine alignment stays near chance in every channel for every model (A=B=C≈ 0.50). A supra-threshold $44\,$px internal-contrast re-run reproduces this result, with A/B/C at 0.51–0.54 and n=41). This is not a hard capability floor (on clean slides the same defect recovers under targeted elicitation once it clears acuity), but on real cluttered layouts the model does not isolate the misaligned element reliably among genuine layout variation, and specificity collapses. For declared geometry, the symbolic linter remains the reliable detector.

Pixel-recovered structure does not restore the linter. A third-party PDF/PNG deck ships no IR, so we ask whether *recovering* element structure from pixels re-arms the linter. Document-layout datasets such as DocLayNet and PubLayNet define region annotations for recovering document structure (Pfitzmann et al. 2022; Zhong, Tang, and Yepes 2019). In our test, PP-DocLayoutV2 returns region boxes per render. Class-agnostic non-maximum suppression removes cross-class duplicates before the boxes feed the *same* linter. The pre-registered result is negative: recovered structure leaves the linter at chance on G1/G3/G4 and rescues only overlap partially (G2 0.54 vs. native-IR 0.67 vs. VLM-only 0.87). The detector *over-segments* ($\approx 2\times$ more boxes, recovery recall at intersection-over-union (IoU)$\geq$0.5 only 0.30–0.39), so collisions split and overlap recall falls $0.78 \rightarrow 0.29$. On image-only SlideAudit the recovered linter clears chance only on G2 (0.55) while the C3-elicited VLM reaches 0.94. On real third-party decks even *native* IR is weakened (no position/font bookkeeping, so G3/G4 rules are at chance and margin/overlap rules *over-fire*, specificity $0.11/0.56$), so the VLM-only engine has higher balanced accuracy on every coarse class in this setting. The linter's advantage is limited to in-distribution, clean native IR. With open-world inputs, the deployable critic remains bounded by the VLM.

8 Limitations

The full critic assumes a generation pipeline with trustworthy native IR. Pixel-only or open-world inputs reduce it to VLM-only inspection, and structure recovered from pixels does not restore the linter's guarantees. The real-layout study uses injected defects, with image-only SlideAudit as a complementary probe. Reported class intervals are component-wise Wilson-derived, but we do not estimate system-level macro uncertainty with templates or decks as clusters. Examiner feedback yields near-zero downstream generation gain: this paper supports failure diagnosis and detection, not the claim that a critic necessarily improves generated slides. The human audit is a corpus spot-check rather than an inter-annotator or deployment-time baseline.

9 Conclusion

Failure attribution guides a manually specified division of labor that is frozen before evaluation: trusted declared geometry goes to symbolic linting, render-only failures to targeted elicitation, and semantics to direct neural inspection. The disjoint evaluation supports this design most strongly for native-IR classes. Neural transfer and incomplete reference comparison remain explicit boundaries. The result applies to IR-owning generation pipelines, not to a universal pixel-only slide critic. *Reproducibility:* code, linters, harness, detailed results, and executor audits are in the Code and Data Supplement.

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